MPG No. 1.1.1 1 [Type I, triclinic] [M tensor]

* Rank 0 tensor.

$$\begin{bmatrix} M \end{bmatrix}$$

$$M = \mathbb{M}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} M_x & M_y & M_z \end{bmatrix}$$

$$M_x = \mathbb{M}_x^{(1)}$$
$$M_y = \mathbb{M}_y^{(1)}$$
$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & M_{xy} & M_{xz} \\ M_{xy} & M_{yy} & M_{yz} \\ M_{xz} & M_{yz} & M_{zz} \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)}$$
$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$
$$M_{xz} = \mathbb{M}_{xz}^{(1)}$$
$$M_{yy} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(1)} - \mathbb{M}_v^{(1)}$$
$$M_{yz} = \mathbb{M}_{yz}^{(1)}$$
$$M_{zz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & M_{xz} \\ -M_{xy} & 0 & M_{yz} \\ -M_{xz} & -M_{yz} & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$
$$M_{xz} = -\mathbb{T}_y^{(1)}$$
$$M_{yz} = \mathbb{T}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & M_{xxy} & M_{xxz} \\ M_{yyx} & M_{yyy} & M_{yyz} \\ M_{zzx} & M_{zzy} & M_{zzz} \\ M_{yzx} & M_{yyz} & M_{zzy} \\ M_{xxz} & M_{yzx} & M_{zzx} \\ M_{xxy} & M_{yyx} & M_{yzx} \end{bmatrix}$$

$$M_{xxx} = 3\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)}$$
$$M_{xxy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\beta(1)}$$
$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\beta(1)}$$
$$M_{yyx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\beta(1)}$$
$$M_{yyy} = 3\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)}$$
$$M_{yyz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\beta(1)}$$
$$M_{zzx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\beta(1)}$$
$$M_{zzy} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\beta(1)}$$
$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$
$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxy} & M_{xxz} \\ M_{yyx} & 0 & M_{yyz} \\ M_{zzx} & M_{zzy} & 0 \\ M_{yzx} & -\frac{M_{yyz}}{2} & -\frac{M_{zzy}}{2} \\ -\frac{M_{xxz}}{2} & M_{zxy} & -\frac{M_{zzx}}{2} \\ -\frac{M_{xxy}}{2} & -\frac{M_{yyx}}{2} & -M_{yzx} - M_{zxy} \end{bmatrix}$$

$$M_{xxy} = 2\mathbb{M}_y^{(2)} - 2\mathbb{T}_{xz}^{(1)}$$
$$M_{xxz} = 2\mathbb{M}_z^{(2)} + 2\mathbb{T}_{xy}^{(1)}$$
$$M_{yyx} = 2\mathbb{M}_x^{(2)} + 2\mathbb{T}_{yz}^{(1)}$$
$$M_{yyz} = 2\mathbb{M}_z^{(2)} - 2\mathbb{T}_{xy}^{(1)}$$
$$M_{zzx} = 2\mathbb{M}_x^{(2)} - 2\mathbb{T}_{yz}^{(1)}$$
$$M_{zzy} = 2\mathbb{M}_y^{(2)} + 2\mathbb{T}_{xz}^{(1)}$$
$$M_{yzx} = 3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$\frac{M_{yyz}}{2} = \mathbb{M}_z^{(2)} - \mathbb{T}_{xy}^{(1)}$$
$$\frac{M_{zzy}}{2} = \mathbb{M}_y^{(2)} + \mathbb{T}_{xz}^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)} + \mathbb{T}_{xy}^{(1)}$$
$$M_{zxy} = -3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$\frac{M_{zzx}}{2} = \mathbb{M}_x^{(2)} - \mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{xxy}}{2} = \mathbb{M}_y^{(2)} - \mathbb{T}_{xz}^{(1)}$$
$$\frac{M_{yyx}}{2} = \mathbb{M}_x^{(2)} + \mathbb{T}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & M_{yzz} \\ M_{zxx} & M_{zxy} & M_{zxz} \\ M_{xyx} & M_{xyy} & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$
$$M_{yzy} = \mathbb{M}_z^{(3)} + \mathbb{T}_{xy}^{(2)}$$
$$M_{yzz} = -\mathbb{M}_y^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{zxx} = -\mathbb{M}_z^{(3)} + \mathbb{T}_{xy}^{(2)}$$
$$M_{zxy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$
$$M_{zxz} = \mathbb{M}_x^{(3)} + \mathbb{T}_{yz}^{(2)}$$
$$M_{xyx} = \mathbb{M}_y^{(3)} + \mathbb{T}_{xz}^{(2)}$$
$$M_{xyy} = -\mathbb{M}_x^{(3)} + \mathbb{T}_{yz}^{(2)}$$
$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & M_{xxyz} & M_{xxzx} & M_{xxxy} \\ M_{xxyy} & M_{yyyy} & M_{yyzz} & M_{yyyz} & M_{yyzx} & M_{yyxy} \\ M_{xxzz} & M_{yyzz} & M_{zzzz} & M_{zzyz} & M_{zzzx} & M_{zzxy} \\ M_{xxyz} & M_{yyyz} & M_{zzyz} & M_{yzyz} & M_{yzzx} & M_{yzxy} \\ M_{xxzx} & M_{yyzx} & M_{zzzx} & M_{yzzx} & M_{zxzx} & M_{zxxy} \\ M_{xxxy} & M_{yyxy} & M_{zzxy} & M_{yzxy} & M_{zxxy} & M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)}$$
$$M_{xxzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)}$$
$$M_{xxyz} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)}$$
$$M_{xxzx} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)}$$
$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{yyyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} - 2\mathbb{M}_v^{(1)} - 4\mathbb{M}_v^{(2)}$$
$$M_{yyzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - \mathbb{M}_v^{(1)}$$
$$M_{yyyz} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)}$$
$$M_{yyzx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)}$$
$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{zzyz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)}$$
$$M_{zzzx} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)}$$
$$M_{yzyz} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - \mathbb{M}_v^{(2)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)}$$
$$M_{yzxy} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)}$$
$$M_{zxzx} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + \mathbb{M}_v^{(2)}$$
$$M_{zxxy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)}$$
$$M_{xyxy} = \mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & M_{xxyz} & M_{xxzx} & M_{xxxy} \\ -M_{xxyy} & 0 & M_{yyzz} & M_{yyyz} & M_{yyzx} & M_{yyxy} \\ -M_{xxzz} & -M_{yyzz} & 0 & M_{zzyz} & M_{zzzx} & M_{zzxy} \\ -M_{xxyz} & -M_{yyyz} & -M_{zzyz} & 0 & M_{yzzx} & M_{yzxy} \\ -M_{xxzx} & -M_{yyzx} & -M_{zzzx} & -M_{yzzx} & 0 & M_{zzxy} \\ -M_{xxxy} & -M_{yyxy} & -M_{zzxy} & -M_{yzxy} & -M_{zzxy} & 0 \end{bmatrix}$$

$$
\begin{aligned}
M_{xxyy} &= -2\mathbb{M}_{v}^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\
M_{xxzz} &= 3\mathbb{M}_{u}^{(3)} - \mathbb{M}_{v}^{(3)} - 4\mathbb{T}_{xyz}^{(1)} \\
M_{xxyz} &= \mathbb{M}_{yz}^{(3)} - 4\mathbb{T}_{x}^{\beta(1)} \\
M_{xxzx} &= \mathbb{M}_{xz}^{(3)} - 2\mathbb{T}_{y}^{(1)} + 2\mathbb{T}_{y}^{\alpha(1)} + 2\mathbb{T}_{y}^{\beta(1)} \\
M_{xxxy} &= \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_{z}^{(1)} - 2\mathbb{T}_{z}^{\alpha(1)} + 2\mathbb{T}_{z}^{\beta(1)} \\
M_{yyzz} &= 3\mathbb{M}_{u}^{(3)} + \mathbb{M}_{v}^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\
M_{yyyz} &= \mathbb{M}_{yz}^{(3)} + 2\mathbb{T}_{x}^{(1)} - 2\mathbb{T}_{x}^{\alpha(1)} + 2\mathbb{T}_{x}^{\beta(1)} \\
M_{yyzx} &= \mathbb{M}_{xz}^{(3)} - 4\mathbb{T}_{y}^{\beta(1)} \\
M_{yyxy} &= \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_{z}^{(1)} + 2\mathbb{T}_{z}^{\alpha(1)} + 2\mathbb{T}_{z}^{\beta(1)} \\
M_{zzyz} &= \mathbb{M}_{yz}^{(3)} - 2\mathbb{T}_{x}^{(1)} + 2\mathbb{T}_{x}^{\alpha(1)} + 2\mathbb{T}_{x}^{\beta(1)} \\
M_{zzzx} &= \mathbb{M}_{xz}^{(3)} + 2\mathbb{T}_{y}^{(1)} - 2\mathbb{T}_{y}^{\alpha(1)} + 2\mathbb{T}_{y}^{\beta(1)} \\
M_{zzxy} &= \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)} \\
M_{yzzx} &= -\mathbb{T}_{z}^{(1)} - 4\mathbb{T}_{z}^{\alpha(1)} \\
M_{yzxy} &= \mathbb{T}_{y}^{(1)} + 4\mathbb{T}_{y}^{\alpha(1)} \\
M_{zxxy} &= -\mathbb{T}_{x}^{(1)} - 4\mathbb{T}_{x}^{\alpha(1)}
\end{aligned}
$$

* Rank 4 tensor (aas).

$$
\begin{bmatrix}
M_{yzyz} & M_{yzzx} & M_{yzxy} \\
M_{yzzx} & M_{zxzx} & M_{zxxy} \\
M_{yzxy} & M_{zxxy} & M_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzyz} &= \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)} + 2\mathbb{M}_{v}^{(4)} \\
M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} \\
M_{yzxy} &= 2\mathbb{M}_{xz}^{(4)} \\
M_{zxzx} &= \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)} - 2\mathbb{M}_{v}^{(4)} \\
M_{zxxy} &= 2\mathbb{M}_{yz}^{(4)} \\
M_{xyxy} &= \mathbb{M}_{0}^{(3)} + 4\mathbb{M}_{u}^{(4)}
\end{aligned}
$$

* Rank 4 tensor (aaa).

$$
\begin{bmatrix}
0 & M_{yzzx} & M_{yzxy} \\
-M_{yzzx} & 0 & M_{zxxy} \\
-M_{yzxy} & -M_{zxxy} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzzx} &= -\mathbb{T}_{z}^{(2)} \\
M_{yzxy} &= \mathbb{T}_{y}^{(2)} \\
M_{zxxy} &= -\mathbb{T}_{x}^{(2)}
\end{aligned}
$$

* Rank 4 tensor (sa).

$$
\begin{bmatrix}
M_{xxyz} & M_{xxzx} & M_{xxxy} \\
M_{yyyz} & M_{yyzx} & M_{yyxy} \\
M_{zzyz} & M_{zzzx} & M_{zzxy} \\
M_{yzyz} & M_{yzzx} & M_{yzxy} \\
M_{zxyz} & M_{zxzx} & M_{zxxy} \\
M_{xyyz} & M_{xyzx} & M_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxyz} &= \mathbb{T}^{(3)}_{x} + 2\mathbb{T}^{(4)}_{x} + 2\mathbb{T}^{\alpha(2)}_{x} \\
M_{xxzx} &= 2\mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(3)}_{y} - \mathbb{T}^{\alpha(2)}_{y} - \mathbb{T}^{\beta(2)}_{y} \\
M_{xxxy} &= -2\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(3)}_{z} - \mathbb{T}^{\alpha(2)}_{z} + \mathbb{T}^{\beta(2)}_{z} \\
M_{yyyz} &= -2\mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(3)}_{x} - \mathbb{T}^{\alpha(2)}_{x} + \mathbb{T}^{\beta(2)}_{x} \\
M_{yyzx} &= \mathbb{T}^{(3)}_{y} + 2\mathbb{T}^{(4)}_{y} + 2\mathbb{T}^{\alpha(2)}_{y} \\
M_{yyxy} &= 2\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(3)}_{z} - \mathbb{T}^{\alpha(2)}_{z} - \mathbb{T}^{\beta(2)}_{z} \\
M_{zzyz} &= 2\mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(3)}_{x} - \mathbb{T}^{\alpha(2)}_{x} - \mathbb{T}^{\beta(2)}_{x} \\
M_{zzzx} &= -2\mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(3)}_{y} - \mathbb{T}^{\alpha(2)}_{y} + \mathbb{T}^{\beta(2)}_{y} \\
M_{zzxy} &= \mathbb{T}^{(3)}_{z} + 2\mathbb{T}^{(4)}_{z} + 2\mathbb{T}^{\alpha(2)}_{z} \\
M_{yzyz} &= -3\mathbb{M}^{(5)}_{u} - \mathbb{M}^{(5)}_{v} + \mathbb{T}^{(2)}_{xyz} \\
M_{yzzx} &= -\mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(4)}_{z} - \mathbb{T}^{\alpha(2)}_{z} - \mathbb{T}^{\beta(2)}_{z} \\
M_{yzxy} &= \mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(4)}_{y} - \mathbb{T}^{\alpha(2)}_{y} + \mathbb{T}^{\beta(2)}_{y} \\
M_{zxyz} &= \mathbb{M}^{(5)}_{xy} + \mathbb{T}^{(4)}_{z} - \mathbb{T}^{\alpha(2)}_{z} + \mathbb{T}^{\beta(2)}_{z} \\
M_{zxzx} &= 3\mathbb{M}^{(5)}_{u} - \mathbb{M}^{(5)}_{v} + \mathbb{T}^{(2)}_{xyz} \\
M_{zxxy} &= -\mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(4)}_{x} - \mathbb{T}^{\alpha(2)}_{x} - \mathbb{T}^{\beta(2)}_{x} \\
M_{xyyz} &= -\mathbb{M}^{(5)}_{xz} + \mathbb{T}^{(4)}_{y} - \mathbb{T}^{\alpha(2)}_{y} - \mathbb{T}^{\beta(2)}_{y} \\
M_{xyzx} &= \mathbb{M}^{(5)}_{yz} + \mathbb{T}^{(4)}_{x} - \mathbb{T}^{\alpha(2)}_{x} + \mathbb{T}^{\beta(2)}_{x} \\
M_{xyxy} &= 2\mathbb{M}^{(5)}_{v} + \mathbb{T}^{(2)}_{xyz}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
M_{yzxx} & M_{yzyy} & M_{yzzz} & M_{yzyz} & M_{yzzx} & M_{yzxy} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & M_{zxyz} & M_{zxzx} & M_{zxxy} \\
M_{xyxx} & M_{xyyy} & M_{xyzz} & M_{xyyz} & M_{xyzx} & M_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzxx} &= \mathbb{T}^{(5)}_{x} + 2\mathbb{T}^{(6)}_{x} + 2\mathbb{T}^{\alpha(3)}_{x} \\
M_{yzyy} &= -2\mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(5)}_{x} - \mathbb{T}^{\alpha(3)}_{x} + \mathbb{T}^{\beta(3)}_{x} \\
M_{yzzz} &= 2\mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(5)}_{x} - \mathbb{T}^{\alpha(3)}_{x} - \mathbb{T}^{\beta(3)}_{x} \\
M_{yzyz} &= -3\mathbb{M}^{(6)}_{u} - \mathbb{M}^{(6)}_{v} + \mathbb{T}^{(3)}_{xyz} \\
M_{yzzx} &= \mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(6)}_{z} - \mathbb{T}^{\alpha(3)}_{z} + \mathbb{T}^{\beta(3)}_{z} \\
M_{yzxy} &= -\mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(6)}_{y} - \mathbb{T}^{\alpha(3)}_{y} - \mathbb{T}^{\beta(3)}_{y} \\
M_{zxxx} &= 2\mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(5)}_{y} - \mathbb{T}^{\alpha(3)}_{y} - \mathbb{T}^{\beta(3)}_{y} \\
M_{zxyy} &= \mathbb{T}^{(5)}_{y} + 2\mathbb{T}^{(6)}_{y} + 2\mathbb{T}^{\alpha(3)}_{y} \\
M_{zxzz} &= -2\mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(5)}_{y} - \mathbb{T}^{\alpha(3)}_{y} + \mathbb{T}^{\beta(3)}_{y} \\
M_{zxyz} &= -\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(6)}_{z} - \mathbb{T}^{\alpha(3)}_{z} - \mathbb{T}^{\beta(3)}_{z} \\
M_{zxzx} &= 3\mathbb{M}^{(6)}_{u} - \mathbb{M}^{(6)}_{v} + \mathbb{T}^{(3)}_{xyz} \\
M_{zxxy} &= \mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(6)}_{x} - \mathbb{T}^{\alpha(3)}_{x} + \mathbb{T}^{\beta(3)}_{x} \\
M_{xyxx} &= -2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(5)}_{z} - \mathbb{T}^{\alpha(3)}_{z} + \mathbb{T}^{\beta(3)}_{z} \\
M_{xyyy} &= 2\mathbb{M}^{(6)}_{xy} + \mathbb{T}^{(5)}_{z} - \mathbb{T}^{\alpha(3)}_{z} - \mathbb{T}^{\beta(3)}_{z} \\
M_{xyzz} &= \mathbb{T}^{(5)}_{z} + 2\mathbb{T}^{(6)}_{z} + 2\mathbb{T}^{\alpha(3)}_{z} \\
M_{xyyz} &= \mathbb{M}^{(6)}_{xz} + \mathbb{T}^{(6)}_{y} - \mathbb{T}^{\alpha(3)}_{y} + \mathbb{T}^{\beta(3)}_{y} \\
M_{xyzx} &= -\mathbb{M}^{(6)}_{yz} + \mathbb{T}^{(6)}_{x} - \mathbb{T}^{\alpha(3)}_{x} - \mathbb{T}^{\beta(3)}_{x} \\
M_{xyxy} &= 2\mathbb{M}^{(6)}_{v} + \mathbb{T}^{(3)}_{xyz}
\end{aligned}
$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
M_{xxxx} & M_{xxyy} & M_{xxzz} & M_{xxyz} & M_{xxzx} & M_{xxxy} & M_{xxzy} & M_{xxxz} & M_{xxyx} \\
M_{yyxx} & M_{yyyy} & M_{yyzz} & M_{yyyz} & M_{yyzx} & M_{yyxy} & M_{yyzy} & M_{yyxz} & M_{yyyx} \\
M_{zzxx} & M_{zzyy} & M_{zzzz} & M_{zzyz} & M_{zzzx} & M_{zzxy} & M_{zzzy} & M_{zzxz} & M_{zzyx} \\
M_{yzxx} & M_{yzyy} & M_{yzzz} & M_{yzyz} & M_{yzzx} & M_{yzxy} & M_{yzzy} & M_{yzxz} & M_{yzyx} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & M_{zxyz} & M_{zxzx} & M_{zxxy} & M_{zxzy} & M_{zxxz} & M_{zxyx} \\
M_{xyxx} & M_{xyyy} & M_{xyzz} & M_{xyyz} & M_{xyzx} & M_{xyxy} & M_{xyzy} & M_{xyxz} & M_{xyyx}
\end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$

$$M_{xxyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)} + \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxyz} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} + \mathbb{T}_x^{(3)} + 2\mathbb{T}_x^{(4)} + 2\mathbb{T}_x^{\alpha(2)} - 4\mathbb{T}_x^{\beta(1)}$$

$$M_{xxzx} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} - 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{xxzy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} - \mathbb{T}_x^{(3)} - 2\mathbb{T}_x^{(4)} - 2\mathbb{T}_x^{\alpha(2)} - 4\mathbb{T}_x^{\beta(1)}$$

$$M_{xxzz} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} - 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{yyxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} + 2\mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{yyyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} - 2\mathbb{M}_v^{(1)} - 4\mathbb{M}_v^{(2)}$$

$$M_{yyzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{yyyz} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} + 2\mathbb{T}_x^{(1)} + \mathbb{T}_x^{(3)} - 2\mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_x^{\beta(2)}$$

$$M_{yyzx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{(4)} + 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)}$$

$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{yyzy} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} + 2\mathbb{T}_x^{(1)} - \mathbb{T}_x^{(3)} - 2\mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} - \mathbb{T}_x^{\beta(2)}$$

$$M_{yyzx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{(4)} - 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)}$$

$$M_{yyyx} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{zzxx} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{zzyy} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)} - \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$

$$M_{zzyz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} - 2\mathbb{T}_x^{(1)} + \mathbb{T}_x^{(3)} + 2\mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} - \mathbb{T}_x^{\beta(2)}$$

$$M_{zzzx} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} + 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$

$$M_{zzzy} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} - 2\mathbb{T}_x^{(1)} - \mathbb{T}_x^{(3)} + 2\mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_x^{\beta(2)}$$

$$M_{zzxz} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} + 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{zzyx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{(4)} - 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$

$$M_{yzxx} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} - \mathbb{M}_{yz}^{(3)} + 4\mathbb{T}_x^{\beta(1)}$$

$$M_{yzyy} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} - 2\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)} - 2\mathbb{T}_x^{\beta(1)}$$

$$M_{yzzz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} + 2\mathbb{T}_x^{(1)} - 2\mathbb{T}_x^{\alpha(1)} - 2\mathbb{T}_x^{\beta(1)}$$

$$M_{yzyz} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)} - \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{yzxy} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(1)} + \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{yzzy} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$

$$M_{yzxz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{yzyx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(1)} - \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{zxxx} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} + 2\mathbb{T}_y^{(1)} - 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)}$$

$$M_{zxyy} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} - \mathbb{M}_{xz}^{(3)} + 4\mathbb{T}_y^{\beta(1)}$$

$$M_{zxzz} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} - 2\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)}$$

$$M_{zxyz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{zxzx} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)} + \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{zxxy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)} - \mathbb{T}_x^{(1)} + \mathbb{T}_x^{(4)} - 4\mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\alpha(2)} - \mathbb{T}_x^{\beta(2)}$$

$$M_{zxzy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{zxxz} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & M_{yzyy} & M_{yzzz} & M_{yzyz} & M_{yzzx} & M_{yzxy} & M_{yzzy} & M_{yzxz} & M_{yzyx} \\ M_{zxxx} & M_{zxyy} & M_{zxzz} & M_{zxyz} & M_{zxzx} & M_{zxxy} & M_{zxzy} & M_{zxxz} & M_{zxyx} \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & M_{xyyz} & M_{xyzx} & M_{xyxy} & M_{xyzy} & M_{xyxz} & M_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
M_{yzxx} &= \mathbb{T}_x^{(5)} + 2\mathbb{T}_x^{(6)} + 2\mathbb{T}_x^{\alpha(3)} \\
M_{yzyy} &= -2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{yzzz} &= 2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{yzyz} &= \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)} + 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzzx} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{yzxy} &= 2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{yzzy} &= -\mathbb{M}_0^{(3)} + 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)} - 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzxz} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{yzyx} &= -2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} - \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zxxx} &= 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zxyy} &= \mathbb{T}_y^{(5)} + 2\mathbb{T}_y^{(6)} + 2\mathbb{T}_y^{\alpha(3)} \\
M_{zxzz} &= -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{zxyz} &= 2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{zxzx} &= \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)} + 3\mathbb{M}_u^{(6)} - 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{zxxy} &= 2\mathbb{M}_{yz}^{(4)} + \mathbb{M}_{yz}^{(6)} - \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{zxzy} &= -2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{zxxz} &= -\mathbb{M}_0^{(3)} + 2\mathbb{M}_u^{(4)} + 3\mathbb{M}_u^{(6)} + 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{zxyx} &= -2\mathbb{M}_{yz}^{(4)} + \mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{xyyy} &= 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{xyzz} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)} \\
M_{xyyz} &= 2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} - \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{xyzx} &= 2\mathbb{M}_{yz}^{(4)} - \mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{xyxy} &= \mathbb{M}_0^{(3)} + 4\mathbb{M}_u^{(4)} + 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{xyzy} &= -2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{xyxz} &= -2\mathbb{M}_{yz}^{(4)} - \mathbb{M}_{yz}^{(6)} - \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{xyyx} &= -\mathbb{M}_0^{(3)} - 4\mathbb{M}_u^{(4)} + 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & M_{xxxy} & M_{xxxz} \\ M_{yyyx} & M_{yyyy} & M_{yyyz} \\ M_{zzzx} & M_{zzzy} & M_{zzzz} \\ M_{yyzx} & M_{yyzy} & M_{yyzz} \\ M_{zzxx} & M_{zzxy} & M_{zzxz} \\ M_{xxyx} & M_{xxyy} & M_{xxyz} \\ M_{yzzx} & M_{yzzy} & M_{yzzz} \\ M_{zxxx} & M_{zxxy} & M_{zxxz} \\ M_{xxyy} + M_{yyzz} - M_{yzzy} - M_{zxxz} + M_{zzxx} & M_{xyyy} & M_{xyyz} \\ -\frac{M_{yyyz}}{2} + \frac{M_{yyzy}}{2} + M_{zxxy} & M_{xyyz} + \frac{M_{zzzz}}{2} - \frac{M_{zzzx}}{2} & \frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} - M_{yyyx} + M_{zzxy} \end{bmatrix}$$

$$M_{xxxx} = 3\mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$
$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)}$$
$$M_{xxxz} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)}$$
$$M_{yyyx} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)}$$
$$M_{yyyy} = 3\mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} - 2\mathbb{M}_v^{(1)} - 4\mathbb{M}_v^{(2)}$$
$$M_{yyyz} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)}$$
$$M_{zzzx} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)}$$
$$M_{zzzy} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)}$$
$$M_{zzzz} = 3\mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{yyzx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)}$$
$$M_{yyzy} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)}$$
$$M_{yyzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)}$$
$$M_{zzxx} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)}$$
$$M_{zzxz} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)}$$
$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 2\mathbb{M}_v^{(3)}$$
$$M_{xxyz} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)}$$
$$M_{yzzy} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)}$$
$$M_{yzzz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)}$$
$$M_{zxxx} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)}$$
$$M_{zxxy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)}$$
$$M_{zxxz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)}$$
$$M_{xyyy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)}$$
$$M_{xyyz} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)}$$